Advanced Trade Studies and Decision Analysis Methods

Fuzzy Theory in Decision Analysis

Fuzziness, Chaos, and Uncertainty

- Think of the most of real-world things handled daily.
  - They are fuzzy and highly uncertain in nature.
  - We need our knowledge and experience to handle them.
  - The methods and/or solutions to handle them are usually highly dynamic (or sometimes chaotic) in nature.

Fuzzy

- Definition:
  - Something blurred or indistinct, in shape or outline (Oxford English Dictionary).
  - Lacking in clarity or definition (Webster).

- Means of specifying how well an object or property satisfies a vague description.
  - "Lack of crispness" considered to be the state of a real-world object or property.

Fuzzy

- Example:

  "John is tall"

  Is this true if John is 5'10"?

  Most people would hesitate to answer "true" or "false" preferring to say, "sort of"

  This is not a question of uncertainty about the external world, as we are sure of John's height.

  The term "tall" does not refer to a sharp demarcation of objects into two classes.

  We consider that there are degrees of tallness.

Fuzzy Set Theory

- L. A. Zadeh, Fuzzy Sets, *Information and Control*, 1965.
  - Describes the degree of belonging of an individual member, which can exist as any real number between 0 and 1, instead of just either 0 or 1.
- Fuzzy Set vs. Crisp Set.
  - "Lack of crispness" is in most cases the truth of a real-world property.
  - Cannot determine exactly (100%) a certain characteristic.
  - All the measured qualities appearing in our world of perception are fuzzy in the sense that it is not the quality itself that is fuzzy, but rather that we cannot always measure these physical qualities with 100% certainty.

Fuzzy Set Theory

- Extension of classic set theory.
  - Classic set theory: elements of fuzzy set theory either are or are not members of the set.
- Example:
  - Set of even natural positive numbers less than 10 is a precise collection of four elements $\{2, 4, 6, 8\}$.
  - The set could be described as a membership function $m_A(x)$ defined over some "universe of discourse," where the function takes on the value 1 when the argument x is an element of set A, and 0 otherwise.
  - A fuzzy set offers the possibility for $m_A(x)$ to take on values other than $\{0, 1\}$, say in the range $(0, \alpha)$, where the value assumed indicates the degree of membership of x in A.
  - For consistency, the possible range of values is often scaled to the real numbers between 0 and 1, inclusive.

Fuzzy Set Theory

- Example:
  - A membership $m_A(x) = 1$ indicates that the element x is definitely a member of A.
  - A membership $m_A(x) = 0$ indicates that the element x is not a member of A.
  - Intermediate values indicate intermediate degrees of membership.
  - This intermediate range represents the extension of fuzzy sets beyond the classic "crisp" sets.

Fuzzy Set Theory

- Back to the example:
  - Treat $Tall$ as a fuzzy predicate.
  - The truth value of $Tall(John)$ is a number between 0 and 1, rather than being just true or false.
- The name "fuzzy set" derives from the interpretation of the predicate as implicitly defining a set of its members.

Fuzzy Set Operations

- A fuzzy set is said to be empty if and only if its membership function is identically 0 on the entire universe of discourse X (all possible arguments to the function).
- Two fuzzy sets A and B are equal, $A = B$, if and only if $m_A(x) = m_B(x)$, for all $x \in X$. This can be written $m_A = m_B$
- The complement of a fuzzy set A is denoted A' and defined as $m_{A'} = 1 - m_A$
- The fuzzy set A is contained in or is a subset of B if and only if $m_A \leq m_B$

Fuzzy Set Operations

- The union of two fuzzy sets A and B is a fuzzy set $C = A \cup B$ with a membership function
  $m_C(x) = \max\{m_A(x), m_B(x)\}, x \in X$
- The intersection of two fuzzy sets A and B is a fuzzy set $C = A \cap B$ with a membership function
  $m_C(x) = \min\{m_A(x), m_B(x)\}, x \in X$

Fuzzy Set Operations

- Example: flower that is yellow and fragrant.
  - $m_{Yellow}(x) = 0.8$
  - $m_{Fragrant}(x) = 0.9$
  - Flower membership in the fuzzy set of yellow and fragrant flowers is $\min\{0.8, 0.9\} = 0.8$
  - Flower membership in the fuzzy set of yellow or fragrant flowers set is $\max\{0.8, 0.9\} = 0.9$
  - Flower membership in the fuzzy set of flowers that are not yellow is $1 - 0.8 = 0.2$

Fuzzy Relationships

- Conditional Statements.
  - If the road is icy, then driving is dangerous.

$$\text{If } A \text{ then } B$$

  - Universe of discourse for A and B need not be the same.
  - Universe of discourse that describes the relationship between A and B is a Cartesian product $A \times B$ of all pairs (a, b) where $a \in A$ and $b \in B$
  - If R is a fuzzy relation from the fuzzy set A to the fuzzy set B, R is a fuzzy subset of $A \times B$, where every pair (a, b) has some associated membership value $m_R(a, b)$

Fuzzy Relationships

- Suppose A = small numbers, over a universe of discourse $U = \{1, 2\}$ and B = large numbers, over a universe of discourse $V = \{1, 2, 3\}$
- Membership values for elements in A are given in ordered pairs $(a, m_A(x))$ as $(1, 1), (2, 0.6)$
- Membership values for elements in B are given as ordered pairs $(b, m_B(x))$ as $(1, 0.2), (2, 0.5), (3, 1)$
- Then $R = A \times B$ is defined to take on a membership that equals the $\min\{m_A(x), m_B(x)\}$ for each pair (a, b)
- R is the fuzzy subset of $U \times V$ that describes the membership of each of its constituent elements (u, v) in terms of the relationship between fuzzy sets A and B

	1	2	3
1	0.2	0.5	1.0
2	0.2	0.5	0.6

Fuzzy Relationships

- Fuzzy relationship can be chained together.
  - If R is a fuzzy relation from X to Y and S is a fuzzy relation from Y to Z, then the composition of fuzzy relations R and S is defined over the pairs $(x, z) \in X \times Z$ as

$$R \circ S = \max_{ij}[\min\{m_R(x, y), m_S(y, z)\}]$$

Fuzzy Control

- By aggregating fuzzy rules it is possible to design complex algorithms for fuzzy control.
- Applications:
  - Maintain an airplane in straight-and-level flight.
  - Hold the temperature of a room at 68 degrees.
  - Keep a car engine at 800 revolutions per minute at idle, even when the air conditioner is turned on or off.
  - Dispatch the elevators in a building to respond to all requests in minimum time.

Fuzzy Logic

- Propositional logic:

P	Q		$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
false	false		true	false	false	true	true
false	true		true	false	true	true	false
true	false		false	false	true	false	false
true	true		false	true	true	true	true

  - $P \Rightarrow Q$ means if P is true then Q is also true; if P is false then nothing is said about Q
  - $P \Leftrightarrow Q$ means P is true if Q is true and P is false if Q is false

Fuzzy Logic

- Method for reasoning with logical expressions describing membership in fuzzy sets.
- Example: $Tall(John) \wedge Heavy(John)$
  - Statement has a fuzzy truth value that is function of the truth value of its components.
- Standard rules for evaluating the fuzzy truth, T, of a complex sentence:

$$T(A \wedge B) = \min(T(A), T(B))$$

$$T(A \vee B) = \max(T(A), T(B))$$

$$T(\neg A) = 1 - T(A)$$

  - Truth-functional system.

Fuzzy Logic

- Truth-functional system example:
  - If

$$T(Tall(Nate)) = 0.6$$
$$T(Heavy(Nate)) = 0.4$$

then,

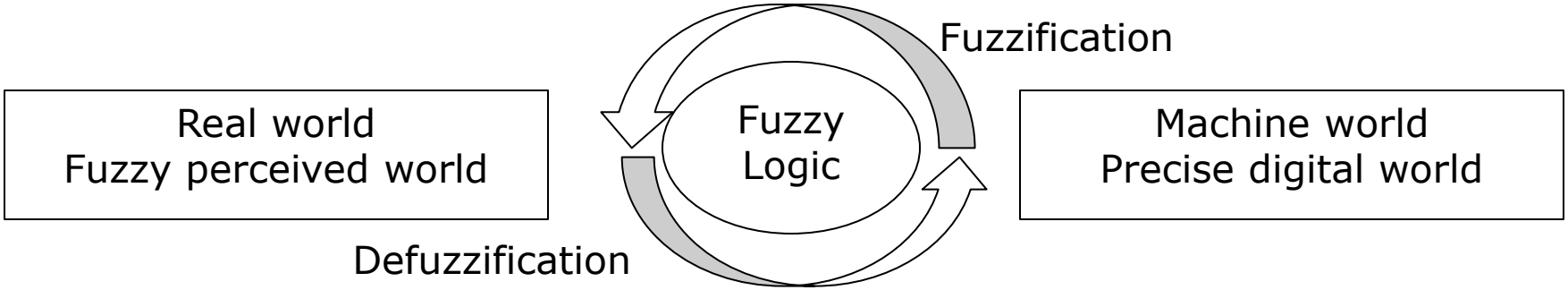
$$T(Tall(Nate)) \wedge T(Heavy(Nate) = 0.4$$ reasonable

$$T(Tall(Nate)) \wedge \neg Tall(Nate) = 0.4$$ not reasonable

Inability of the truth-functional approach to take into account the correlations or anti-correlations among the component properties.

Fuzzy Logic

- Reflects the reality of our world of experience (our perceived world), which is uncertain and fuzzy.
- Convert the "precise" values and measurements in the machine world (digital world) to the "imprecise" values in the real world (perceived world) which is full of fuzziness and uncertainty.
  - Forward operation is called fuzzification scheme.
  - Reverse operation is called defuzzification scheme.



Fuzzy Logic

- Can be decomposed into four basic components.
  - ICMR structural framework of fuzzy logic.

- Fuzzy Identification Module (FIM).
- Fuzzy Categorization Module (FCM).
- Fuzzy Modeling Module (FMM).
- Fuzzy Reasoning Module (FRM).

Fuzzy Identification Module

- Determination and identification of the fuzzy variable (FV) being used by the system.
- Statement: "John is tall"
- FV = "height"
  - The quality that describes one of John's physical characteristics.
  - Does not appear explicitly in the statement.
  - Inferred using common sense or prior knowledge.
- Statement: "Most students in the class are tall"
- FV = "most"
- It is not the quality "height" as before.
  - The quality that describes the proportion of students that are tall in the class.

Fuzzy Characterization Module

- Categorize the chosen $FV(s)$ into different fuzzy sets.
- FV = "height"
- Possible fuzzy sets having a range of values:
  - "short" – for student height below 1.60 m
  - "average" – for student height between 1.55 m and 1.70 m
  - "tall" – for student height over 1.65 m
- Fuzziness in the boundaries.
  - No strict criteria to decide the number, type, and nature of the fuzzy sets.
  - No clear boundary between one fuzzy set and another, which is totally unacceptable in a traditional crisp set.
  - Possible fuzzy sets: "very short", "short", "average", "tall", "very tall"

Fuzzy Modeling Module

- Define the Membership Function (MF) for each fuzzy set.
- MP defines the fuzzy mapping – the fuzzy belonging of each element to its fuzzy set.
- Let X be a collection of objects denoted by

$$\{x \mid x \in X\}$$

  - Called universe of discourse.
- Fuzzy set A in this universe of discourse X is characterized by a membership function $\mu_A(x)$ which takes values in the interval $[0, 1]$.

$$A = \{(x, \mu_A(x) \mid x \in X)\}$$

Fuzzy Modeling Module

- If X is continuous, the fuzzy set is written as:

$$A = \int_X \mu_A(x)dx$$

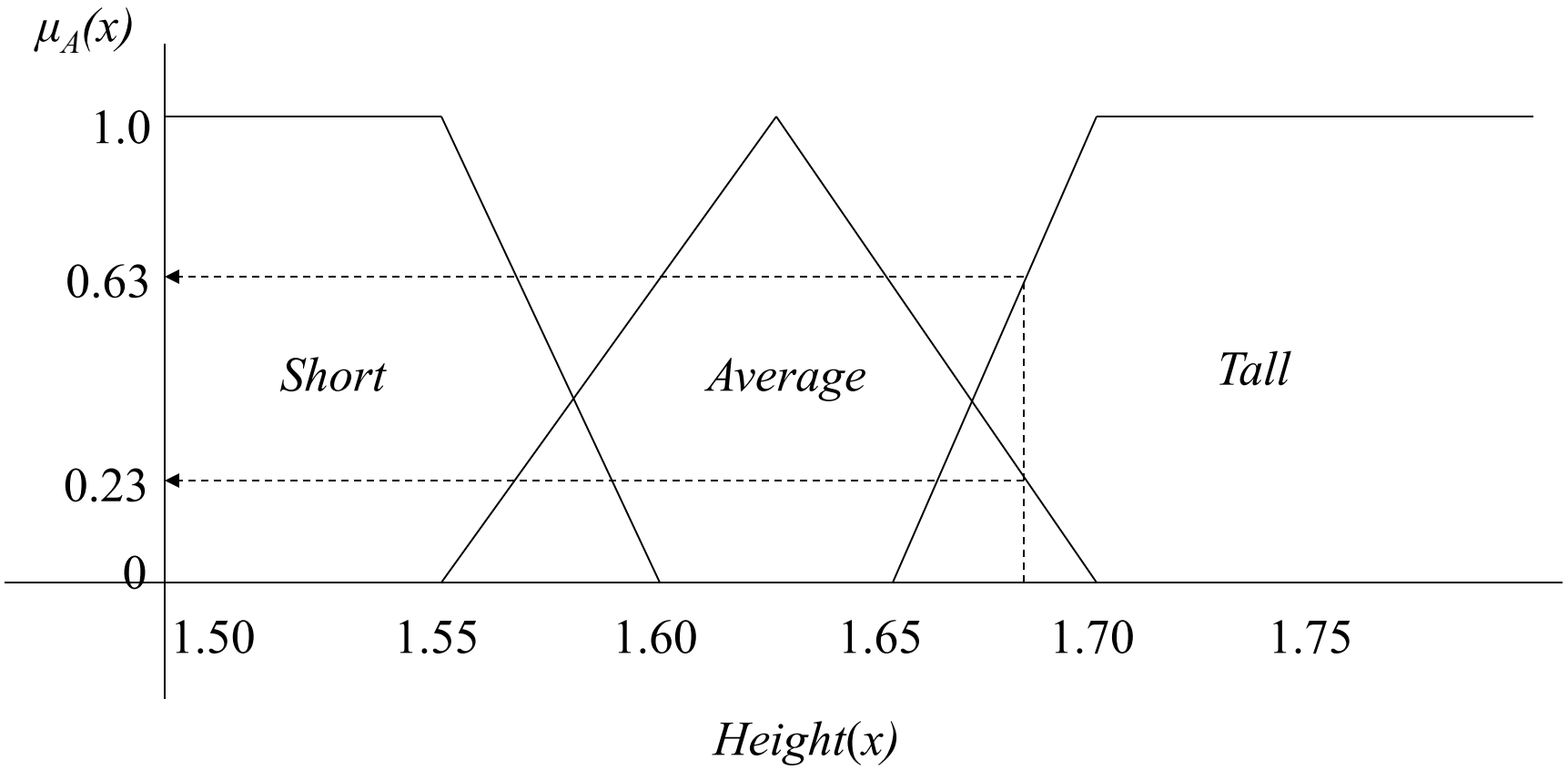
- If X is discrete, the fuzzy set is written as:

$$A = \sum_N \mu_A(x_i) / x_i$$

  - N is the total number of possible discrete values of A.

Fuzzy Modeling Module

- Fuzzy Membership Functions.



Fuzzy Modeling Module

- Example: "the height of student(Jack) is 1.68"
- Membership functions for fuzzy sets will be:
  - "Average" $\mu_A(x) = 0.23$
  - "Tall" $\mu_A(x) = 0.63$
  - Jack is tall, but he is not among the tallest one in the class.
  - Overall speaking, Jack is taller than the average students in the class.

Fuzzy Reasoning Module

- Computational component that aims at the exploration of the heuristics of the system.
- Apply the fuzzy reasoning technique.

- Three main processes:
  - Fuzzy Rules Construction Process (FRCP).
  - Fuzzy Knowledge Base Process (FKBP).
  - Fuzzy Inference Process (FIP).

Fuzzy Rules Construction Process

- Fuzzy rules are sets of fuzzy relations (FR), which normally appear in the form of "if-then" propositions.
  - Also called fuzzy implications functions (FIF).
- If x is A, then y is B
  - where $x, A \in X$ and $y, B \in Y$
- Or it can be written as: $R = A \rightarrow B$
  - where R is the fuzzy rule.

Fuzzy Knowledge Base Process

- The fuzzy knowledge base if the knowledge center of the fuzzy system.
- Collection of fuzzy rules to represent a particular domain of knowledge.

$$R_k = A_k \rightarrow B_k, k \in [1 \ldots N]$$

  - where N is the total number of fuzzy rules.

- Various approaches to implement the fuzzy relation operations.

Fuzzy Knowledge Base Process

- Fuzzy rules implementation schemes.

Method Used	Fuzzy Rule "If x is A then y is B" where $x, A \in X$, $y, B \in Y$
Mini Rule	$R_{Mini} = A \times B = \int_{X \times Y} \mu_A(x) \wedge \mu_B(y) / (x, y)$
Product Rule	$R_{product} = A \times B = \int_{X \times Y} \mu_A(x) \bullet \mu_B(y) / (x, y)$
Max-Min Rule	$R_{Max-Min} = (A \times B) \cup (\neg A \times Y) = \int_{X \times Y} (\mu_A(x) \wedge \mu_B(y)) \vee (1 - \mu_A(x)) / (x, y)$
Arithmetic Rule	$R_{Arithmetic} = (\neg A \times Y) \oplus (X \times B) = \int_{X \times Y} 1 \wedge (1 - \mu_A(x) + \mu_B(y)) / (x, y)$

Fuzzy Inference Process

- Fuzzy inference refers to the fuzzy reasoning operations based on the knowledge in the fuzzy system.
  - Given two groups of fuzzy sets $A, A' \in X$ and $B, B' \in Y$
  - Direct Reasoning:
    - For any two variables x and y:
    - Implication: If x is A, then y is B
    - Premise: x *is* A'
    - Conclusion: y is B'
    - This can be denoted as: $B' = A' \circ R$ (where R is the relation)
  - Indirect Reasoning:
    - For any two variables x and y:
    - Implication: If x is A, then y is B
    - Premise: y *is* B'
    - Conclusion: x is A'
    - This can be denoted as: $A' = R \circ B'$ (where R is the relation)

Fuzzy Inference Process

- Fuzzy compositional rule for fuzzy inference.
  - When there is a complex knowledge base containing a list of fuzzy rules (R_k) with the corresponding input signals A'_k for each fuzzy set A_k.

$$X = (A'_1, A'_2, \ldots, A'_k, \ldots, A'_n)$$
$$Y = B'_j, j = [1 \ldots n]$$

  - Using direct reasoning, the fuzzy variable B' can be inferred as the composition operation between fuzzy set A_k and the fuzzy relation R, as follows:

$$B' = (A'_1, A'_2, \ldots, A'_k, \ldots, A'n) \circ R$$

Fuzzy Inference Process

- Fuzzy compositional rule for fuzzy inference.
  - Two composition schemes are commonly used.

    Sup-min operation.

    Sup-product operation.

  - Composite fuzzy reasoning consists of the following steps:

    1. Identify each fuzzy rule and determine the "connectives" for each condition in the rule base, in which an "OR" clause represents a union operation and an "AND" clause represents an intersection operation.

    2. For each fuzzy rule (R_k), locate the α_k.

    3. Using either the sup-min operation or sup-product operation method, infer the overall defuzzification control action.

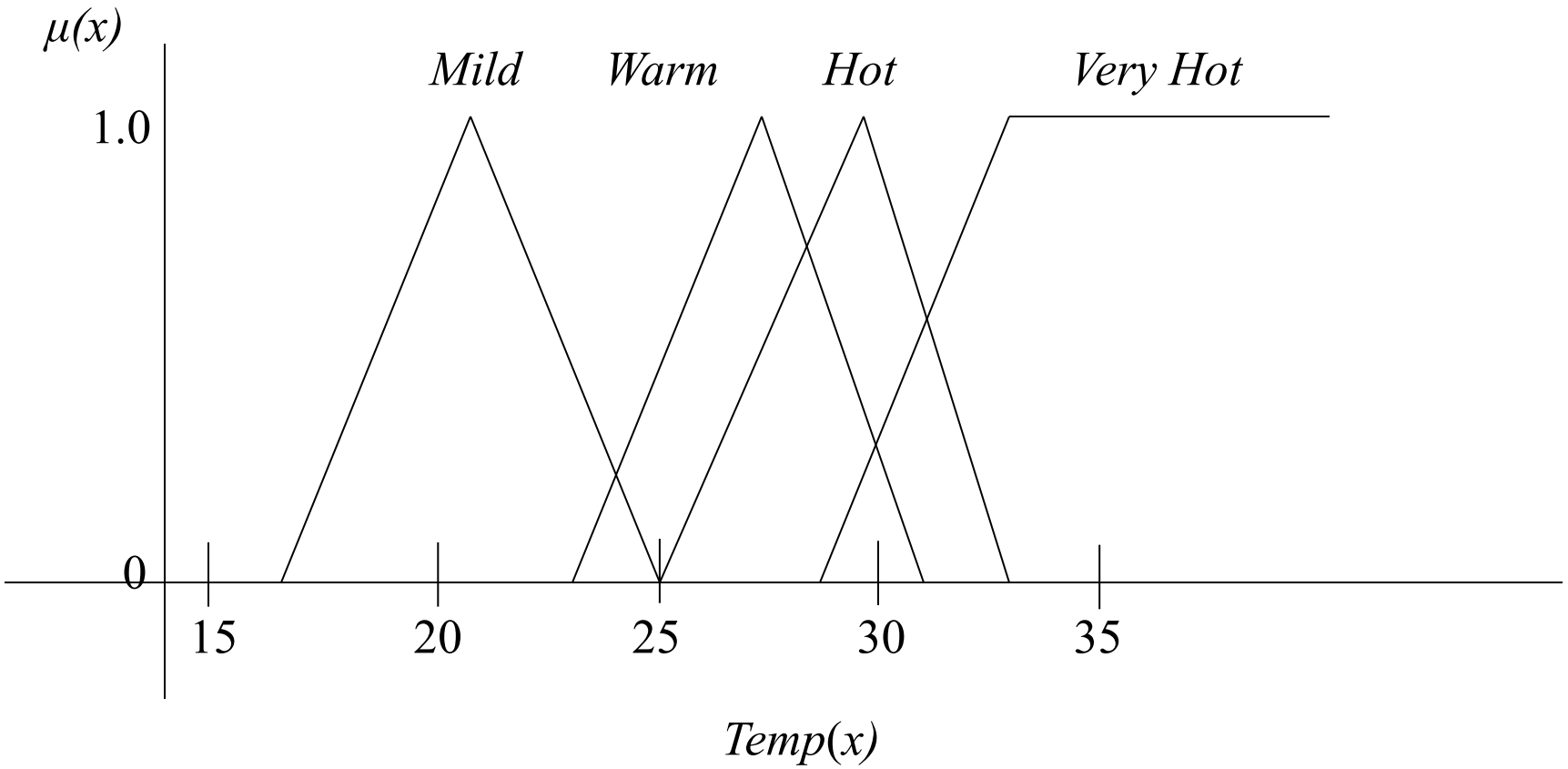
Fuzzy Air-Conditioning System

- System can vary its power control ($Power$) by two environmental factors of the surrounding air condition, namely air temperature ($Temp$) and relative humidity (RH), which are given by two sets of fuzzy variables:
  - $Temp = \{Very\ Cold, Cold, Cool, Mild, Warm, Hot, Very\ Hot\}$
  - $RH = \{Very\ Dry, Dry, Moderate, Humid, Very\ Humid\}$

Temp description (fuzzy variables)	Fuzzy *Temp* range (°C)	*RH* (fuzzy variables)	Fuzzy *RH* range (%)
Mild	17-25	*Dry*	≤ 60
Warm	23-31	*Moderate*	55-85
Hot	25-33	*Humid*	65-95
Very Hot	≥ 28	*Very Humid*	> 75

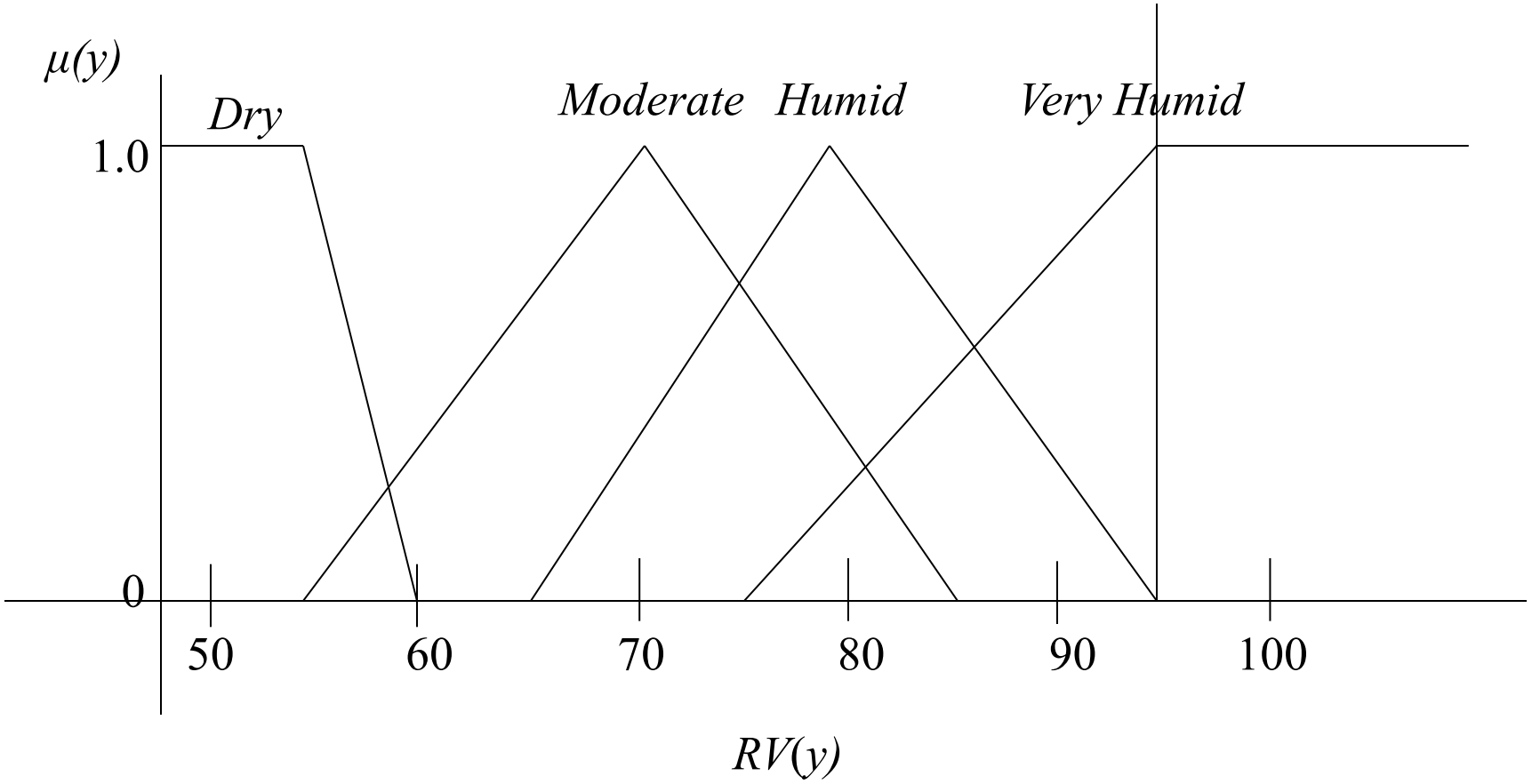
Fuzzy Air-Conditioning System

- Fuzzy Membership Functions - *Temp*



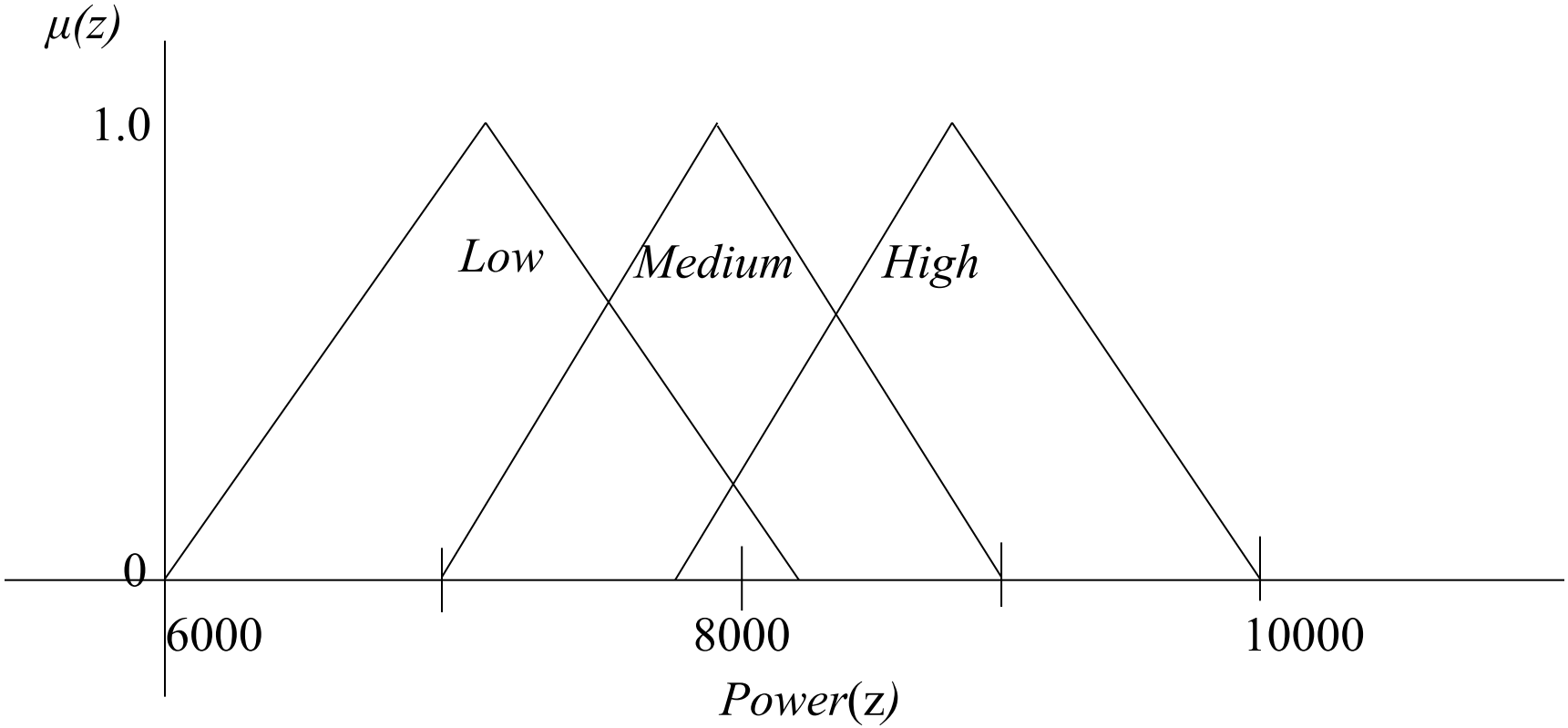
Fuzzy Air-Conditioning System

- Fuzzy Membership Functions - *RV*



Fuzzy Air-Conditioning System

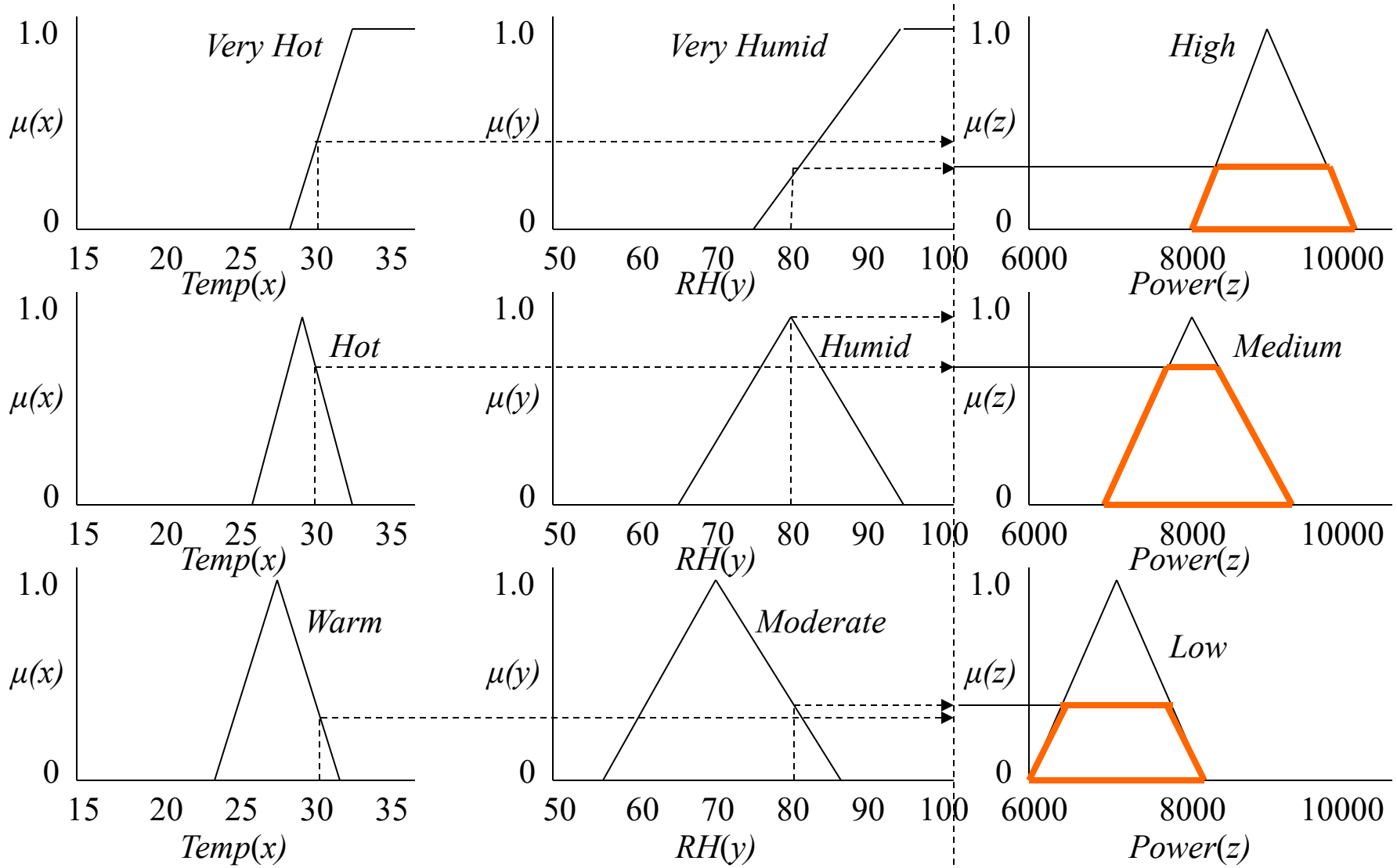
- Power has three different fuzzy states:
  - $Power = \{Low, Medium, High\}$
- Fuzzy Membership Functions - *Power*



Fuzzy Air-Conditioning System

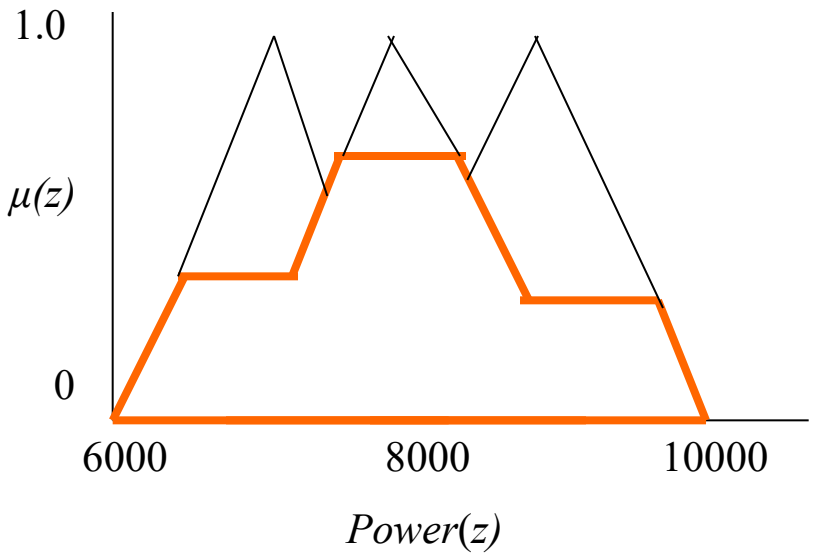
- The power switch is controlled by the following fuzzy rules:
  - Rule 1: If *Temp* is *"Very Hot"* **AND** *RH* is *"Very Humid"*, then switch to *"High"*
  - Rule 2: If *Temp* is *"Hot"* **AND** *RH* is *"Humid"*, then switch to *"Medium"*
  - Rule 3: If *Temp* is *"Warm"* **OR** *RH* is *"Moderate"*, then switch to *"Low"*
- Let $Temp = 30°\mathrm{C}$ and $RH = 80\%$
  - Sup-min method defuzzification scheme.
  - Sup-product method defuzzification scheme.

Sup-Min Operation Method

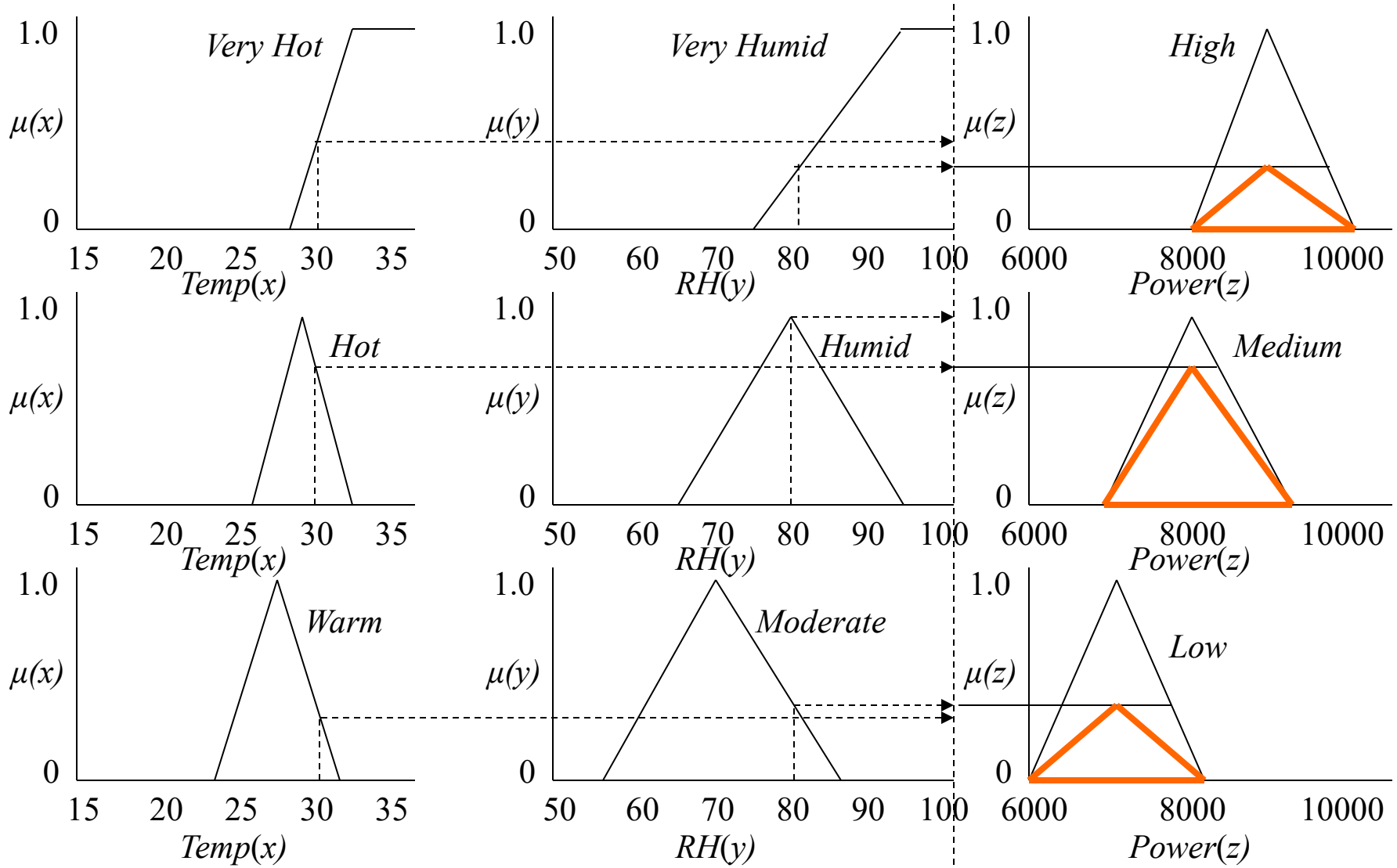


Sup-Min Operation Method

- Control Action Rules:

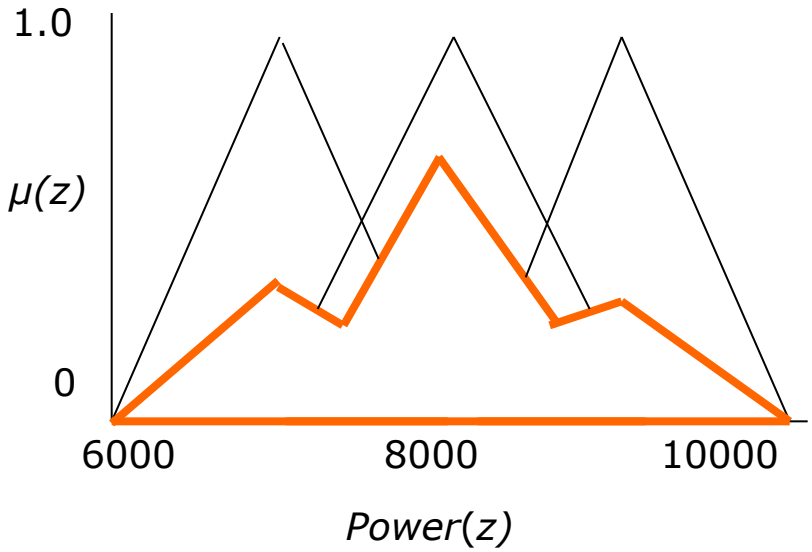


Sup-Product Operation Method



Sup-Product Operation Method

- Control Action Rules:



Fuzzy Systems Applications

- Fuzzy electronic appliances.
  - Fuzzy cookers, fuzzy air-conditioners, fuzzy washing-machines.
- Control systems.
  - Fuzzy robotics, fuzzy automated braking systems.
  - Fuzzy systems for active suspension, automatic transmission, engine control emissions.

- Fuzzy expert system.
- Integration of fuzzy logic with expert knowledge in classical expert systems.
  - Fuzzy systems can handle imprecise concepts and values.

Fuzzy-Neuro Systems

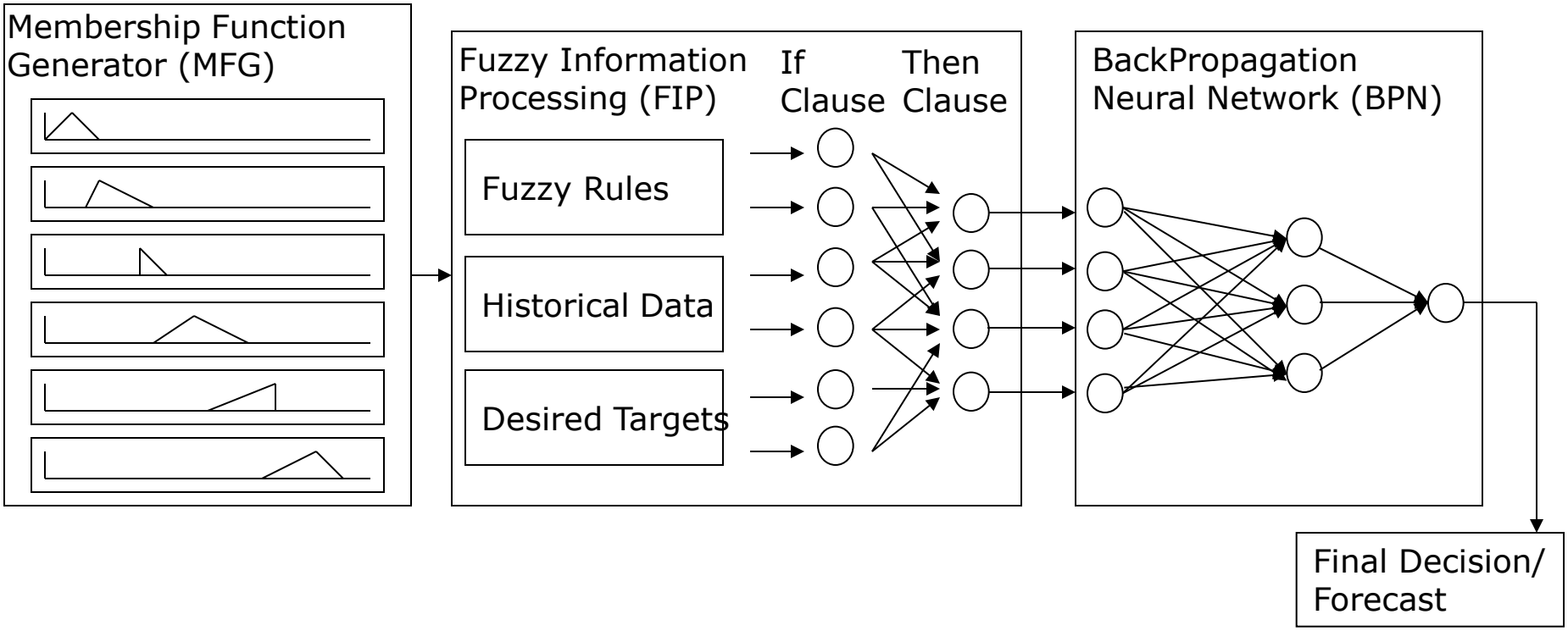
- Major drawback of fuzzy systems is their lack of learning capabilities.
- Solution:
  - Hybridization with other computational intelligence models that provide automatic machine learning.
  - Ex: Neural networks.
- Outcome:
  - Fuzzy-Neuro Systems.
  - Used for solving problems such as stock prediction, severe weather forecasting, robot vision.

FuzzyNet Model

- Consists of:
  - Membership Function Generator (MFG).
    - Provided by domain experts, or automatically generated by using some statistical methods based on historical data.
  - Fuzzy Information Processor (FIP).
    - Accepts three types of information from the database, fuzzy rules and initial weights indicating the "credibility" of rules, historical data, and current data.
  - Back-Propagation Neural Network (BPN).
    - Similar to conventional BPNs.
    - Processing elements used are neural gates generated by the FIP module.

FuzzyNet Model

- Schematic Diagram



Using Fuzzy Logic

- When the inputs or model parameters are uncertain but not random.
- Fuzzy numbers have a range; the larger the range, the more uncertainty.
- Fuzzy numbers have a degree of belief (membership function) between 0 and 1 associated with each point in the range.
  - The degree of belief is not a probability of occurrence.
  - The degree of belief curve does not integrate to 1, and integrating it does not produce a CDF.

Using Fuzzy Logic

- Triangular fuzzy number $<l, m, h>$

- Degree of belief is 0 from $-\infty$ to 1 and from h to ∞.
- Degree of belief at m is 1, from which the degree of belief varies linearly to 0 at both l and h.
- Number "5" can be represented as: $<3, 5, 9>$, $<4, 5, 6>$, $<5, 5, 5.5>$, or $<5, 5, 5>$ depending on the potential values the number might have other than exactly 5.

Using Fuzzy Logic

- Triangular fuzzy number $<l, m, h>$

$$a \times <l, m, h> + b = <al + b, am + b, ah + b>$$
$$<l_1, m_1, h_1> + <l_2, m_2, h_2> = <l_1 + l_2, m_1 + m_2, h_1 + h_2>$$
$$<l_1, m_1, h_1> \times <l_2, m_2, h_2> = <l_1 \times l_2, m_1 \times m_2, h_1 \times h_2>$$

- When a deterministic number is needed to implement the result of an analysis, the fuzzy output must be defuzzified.
- An appropriate deterministic representation is considered the centroid of the fuzzy range.
- Defuzzified value for $<l, m, h>$ is $(l + m + h)/3$

Using Fuzzy Logic for Prediction

- Develop a new system.
  - Both developer and customer want to estimate the cost and schedule with some margin to compensate for the risks associated with making the system operational.
  - If some aspect of the system has never been done before, both the cost and the time to complete are uncertain, but not random.
  - Many times, probability theory is used to estimate the risk-adjusted cost and schedule.

Using Fuzzy Logic for Prediction

- Factors responsible for uncertainty.
- Item technical maturity: is it something done before or just an idea?
  - Define a scale to quantify an item's technical maturity, at least in a fuzzy sense.
- Certainty with the estimate, named proposal risk.
  - Define a scale to enable it to be quantified.
- Cost margin is determined based on fuzzy rules concerning technical maturity and proposal risk.

Using Fuzzy Logic for Prediction

- Identify the fuzzy technical maturity level.

Technical Maturity Definition	Technical Maturity Level
Successfully operating	8
Qualified but not yet used operationally	7
Prototype demonstrated in operational environment	6
Prototype demonstrated in relevant environment	5
Portions validated by analysis or in laboratory	4
Analysis or experiments show concept could work	3
Concept defined	2
Solution scientifically feasible	1
No known approach to meet requirements	0

Using Fuzzy Logic for Prediction

- Identify the fuzzy technical maturity level.
  - Suppose the system needs a new propellant tank with twice the volume as any previously made tank, but to be built exactly the same way as the smaller tanks that have been successfully operated.
    - Item maturity level is less than 8.
    - Item has a known approach to meet requirements, so maturity level is more than 0.
    - Consider a degree of belief related to the technical maturity <3, 5, 7>
    - If analysis can prove that the item meets its requirements (maturity level of 3).
    - If consider that previously smaller items were qualified and successfully operated (maturity level of 7).
    - If consider that the previous smaller items are effectively the prototype for the new item (maturity level of 5).

Using Fuzzy Logic for Prediction

- Identify the fuzzy technical maturity level.

Fuzzy Technical Maturity Definition	Fuzzy Technical Maturity Level
Very mature	<6, 8, 8>
Mature	<3, 5, 7>
Immature	<1, 2, 4>
Very immature	<0, 0, 2>

Using Fuzzy Logic for Prediction

- Identify the fuzzy cost estimates.

Cost Estimate Certainty Definition	Cost Estimate Certainty Level
No prior similar efforts and for many reasons actual cost may be different than predicted	0
Mostly never done before with more cost uncertainties than certainties	1
Similar to prior efforts with reasonable certainty that costs can be predicted accurately	2
Nearly identical to prior efforts and costs are highly predictable	3

Using Fuzzy Logic for Prediction

- Identify the fuzzy cost estimates.
  - If we are very certain our estimate has little certainty, the proposal risk can be categorized as <1, 1, 1>
  - If we have some fear our estimate is off, the proposal risk can be categorized as <1, 2, 3>

Using Fuzzy Logic for Prediction

- Identify the fuzzy cost estimates.

Fuzzy Proposal Risk Definition	Cost Uncertainty Level
Low	<1, 2, 3>
Medium	<0.5, 1.5, 2.5>
High	<0, 0, 2>

Using Fuzzy Logic for Prediction

- Define fuzzy rules.

  IF Technical_Maturity_Level AND Proposal_Risk_Level, THEN Margin_Recommended

Proposal Risk Level	High	<0.25, 0.75, 1.50>	<0.25, 0.50, 1.00>	<0.10, 0.25, 0.75>	<0.00, 0.15, 0.25>
	Medium	<0.25, 0.50, 0.75>	<0.10, 0.25, 0.75>	<0.00, 0.15, 0.50>	<-0.05, 0.05, 0.25>
	Low	<0.10, 0.25, 0.75>	<0.00, 0.15, 0.25>	<-0.10, 0.05, 0.25>	<-0.10, 0.00, 0.15>
Fuzzy Margin		Very Immature	Immature	Mature	Very Mature
		Technical Maturity Level			

Using Fuzzy Logic for Prediction

- Calculate fuzzy cost estimate.

CWBS	Technical maturity	Cost estimate ($K)	Cost certainty	Cost estimate with margin ($K)
1	<6, 7, 8>	1500	<1, 3, 3>	1550
2	<3, 5, 8>	200	<1, 2, 3>	228
3	<1, 3, 5>	2000	<0, 1, 2>	2950
4	<7, 7, 7>	500	<2, 3, 3>	508
Total	-	4200	-	5237